



2118 FAMILY

16,384 x 1 BIT DYNAMIC RAM

	2118-10	2118-12	2118-15
Maximum Access Time (ns)	100	120	150
Read, Write Cycle (ns)	235	270	320
Read-Modify-Write Cycle (ns)	285	320	410

- Single +5V Supply, $\pm 10\%$ Tolerance
- HMOS Technology
- Low Power: 150 mW Max. Operating
11 mW Max. Standby
- Low V_{DD} Current Transients
- All Inputs, Including Clocks,
TTL Compatible
- $\overline{CAS}$ Controlled Output is
Three-State, TTL Compatible
- $\overline{RAS}$ Only Refresh
- 128 Refresh Cycles Required
Every 2ms
- Page Mode and Hidden
Refresh Capability
- Allows Negative Overshoot
 $V_{IL} \text{ min} = -2V$

The Intel® 2118 is a 16,384 word by 1-bit Dynamic MOS RAM designed to operate from a single +5V power supply. The 2118 is fabricated using HMOS — a production proven process for high performance, high reliability, and high storage density.

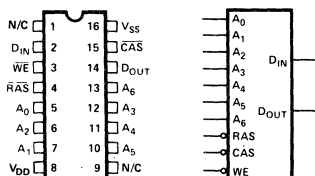
The 2118 uses a single transistor dynamic storage cell and advanced dynamic circuitry to achieve high speed with low power dissipation. The circuit design minimizes the current transients typical of dynamic RAM operation. These low current transients contribute to the high noise immunity of the 2118 in a system environment.

Multiplexing the 14 address bits into the 7 address input pins allows the 2118 to be packaged in the industry standard 16-pin DIP. The two 7-bit address words are latched into the 2118 by the two TTL clocks, Row Address Strobe ($\overline{RAS}$) and Column Address Strobe ($\overline{CAS}$). Non-critical timing requirements for $\overline{RAS}$ and $\overline{CAS}$ allow use of the address multiplexing technique while maintaining high performance.

The 2118 three-state output is controlled by $\overline{CAS}$, independent of $\overline{RAS}$. After a valid read or read-modify-write cycle, data is latched on the output by holding $\overline{CAS}$ low. The data out pin is returned to the high impedance state by returning $\overline{CAS}$ to a high state. The 2118 hidden refresh feature allows $\overline{CAS}$ to be held low to maintain latched data while $\overline{RAS}$ is used to execute $\overline{RAS}$ -only refresh cycles.

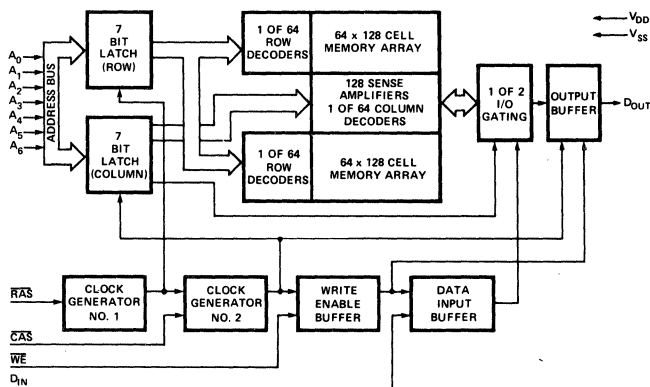
The single transistor storage cell requires refreshing for data retention. Refreshing is accomplished by performing $\overline{RAS}$ -only refresh cycles, hidden refresh cycles, or normal read or write cycles on the 128 address combinations of A_0 through A_6 during a 2ms period. A write cycle will refresh stored data on all bits of the selected row except the bit which is addressed.

PIN
CONFIGURATION LOGIC SYMBOL



A_0 - A_6	ADDRESS INPUTS
$\overline{CAS}$	COLUMN ADDRESS STROBE
D_{IN}	DATA IN
D_{OUT}	DATA OUT
$\overline{WE}$	WRITE ENABLE
$\overline{RAS}$	ROW ADDRESS STROBE
V_{DD}	POWER (+5V)
V_{SS}	GROUND

BLOCK DIAGRAM



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ABSOLUTE MAXIMUM RATINGS*

Ambient Temperature Under Bias	-10°C to +80°C
Storage Temperature	-65°C to +150°C
Voltage on Any Pin Relative to V _{SS}	7.5V
Data Out Current	50mA
Power Dissipation	1.0W

*COMMENT:

Stresses above those listed under "Absolute Maximum Rating" may cause permanent damage to the device. This is a stress rating only and functional operation of the device at these or at any other condition above those indicated in the operational sections of this specification is not implied. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.

D.C. AND OPERATING CHARACTERISTICS^[1]

T_A = 0°C to 70°C, V_{DD} = 5V ±10%, V_{SS} = 0V, unless otherwise noted.

Symbol	Parameter	Limits			Unit	Test Conditions	Notes
		Min.	Typ. ^[2]	Max.			
I _{LI}	Input Load Current (any input)		0.1	10	μA	V _{IN} =V _{SS} to V _{DD}	
I _{LO}	Output Leakage Current for High Impedance State		0.1	10	μA	Chip Deselected: $\overline{\text{CAS}}$ at V _{IH} , V _{OUT} = 0 to 5.5V	
I _{DD1}	V _{DD} Supply Current, Standby		1.2	2	mA	$\overline{\text{CAS}}$ and $\overline{\text{RAS}}$ at V _{IH}	
I _{DD2}	V _{DD} Supply Current, Operating		23	27	mA	2118-10, t _{RC} = t _{RCMIN}	3
			21	25	mA	2118-12, t _{RC} = t _{RCMIN}	3
			19	23	mA	2118-15, t _{RC} = t _{RCMIN}	3
I _{DD3}	V _{DD} Supply Current; $\overline{\text{RAS}}$ -Only Cycle		16	18	mA	2118-10, t _{RC} = t _{RCMIN}	3
			14	16	mA	2118-12, t _{RC} = t _{RCMIN}	3
			12	14	mA	2118-15, t _{RC} = t _{RCMIN}	3
I _{DD5}	V _{DD} Supply Current, Standby, Output Enabled		2	4	mA	$\overline{\text{CAS}}$ at V _{IL} , $\overline{\text{RAS}}$ at V _{IH}	3
V _{IL}	Input Low Voltage (all inputs)	-2.0		0.8	V		
V _{IH}	Input High Voltage (all inputs)	2.4		7.0	V		
V _{OL}	Output Low Voltage			0.4	V	I _{OL} = 4.2mA	
V _{OH}	Output High Voltage	2.4			V	I _{OH} = -5mA	

NOTES:

1. All voltages referenced to V_{SS}.
2. Typical values are for T_A = 25°C and nominal supply voltages.
3. I_{DD} is dependent on output loading when the device output is selected. Specified I_{DD} MAX is measured with the output open.

CAPACITANCE^[1]

T_A = 25°C, V_{DD} = 5V ±10%, V_{SS} = 0V, unless otherwise noted.

Symbol	Parameter	Typ.	Max.	Unit
C _{I1}	Address, Data In	3	5	pF
C _{I2}	$\overline{\text{RAS}}$, $\overline{\text{CAS}}$, $\overline{\text{WE}}$, Data Out	4	7	pF

NOTES:

1. Capacitance measured with Boonton Meter or effective capacitance calculated from the equation:

$$C = \frac{I_{\Delta V}}{\Delta V} \text{ with } \Delta V \text{ equal to 3 volts and power supplies at nominal levels.}$$

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A.C. CHARACTERISTICS^[1,2,3]

T_A = 0°C to 70°C, V_{DD} = 5V ±10%, V_{SS} = 0V, unless otherwise noted.

READ, WRITE, READ-MODIFY-WRITE AND REFRESH CYCLES

Symbol	Parameter	2118-10		2118-12		2118-15		Unit	Notes
		Min.	Max.	Min.	Max.	Min.	Max.		
t _{RAC}	Access Time From $\overline{\text{RAS}}$		100		120		150	ns	4,5
t _{CAC}	Access Time From $\overline{\text{CAS}}$		55		65		80	ns	4,5,6
t _{REF}	Time Between Refresh		2		2		2	ms	
t _{RP}	$\overline{\text{RAS}}$ Precharge Time	110		120		135		ns	
t _{CPN}	$\overline{\text{CAS}}$ Precharge Time (non-page cycles)	50		55		70		ns	
t _{CRP}	$\overline{\text{CAS}}$ to $\overline{\text{RAS}}$ Precharge Time	0		0		0		ns	
t _{RCD}	$\overline{\text{RAS}}$ to $\overline{\text{CAS}}$ Delay Time	25	45	25	55	25	70	ns	7
t _{RSH}	$\overline{\text{RAS}}$ Hold Time	70		85		105		ns	
t _{CSH}	$\overline{\text{CAS}}$ Hold Time	100		120		165		ns	
t _{ASR}	Row Address Set-Up Time	0		0		0		ns	
t _{RAH}	Row Address Hold Time	15		15		15		ns	
t _{ASC}	Column Address Set-Up Time	0		0		0		ns	
t _{CAH}	Column Address Hold Time	15		15		20		ns	
t _{AR}	Column Address Hold Time, to $\overline{\text{RAS}}$	60		70		90		ns	
t _T	Transition Time (Rise and Fall)	3	50	3	50	3	50	ns	8
t _{OFF}	Output Buffer Turn Off Delay	0	45	0	50	0	60	ns	

READ AND REFRESH CYCLES

t _{RC}	Random Read Cycle Time	235		270		320		ns	
t _{RAS}	$\overline{\text{RAS}}$ Pulse Width	115	10000	140	10000	175	10000	ns	
t _{CAS}	$\overline{\text{CAS}}$ Pulse Width	55	10000	65	10000	95	10000	ns	
t _{RCS}	Read Command Set-Up Time	0		0		0		ns	
t _{RCH}	Read Command Hold Time	0		0		0		ns	

WRITE CYCLE

t _{RC}	Random Write Cycle Time	235		270		320		ns	
t _{RAS}	$\overline{\text{RAS}}$ Pulse Width	115	10000	140	10000	175	10000	ns	
t _{CAS}	$\overline{\text{CAS}}$ Pulse Width	55	10000	65	10000	95	10000	ns	
t _{WCS}	Write Command Set-Up Time	0		0		0		ns	9
t _{WCH}	Write Command Hold Time	25		30		45		ns	
t _{WCR}	Write Command Hold Time, to $\overline{\text{RAS}}$	70		85		115		ns	
t _{WP}	Write Command Pulse Width	25		30		50		ns	
t _{RWL}	Write Command to $\overline{\text{RAS}}$ Lead Time	60		65		110		ns	
t _{CWL}	Write Command to $\overline{\text{CAS}}$ Lead Time	45		50		100		ns	
t _{DS}	Data-In Set-Up Time	0		0		0		ns	
t _{DH}	Data-In Hold Time	25		30		45		ns	
t _{DHR}	Data-In Hold Time, to $\overline{\text{RAS}}$	70		85		115		ns	

READ-MODIFY-WRITE CYCLE

t _{RWC}	Read-Modify-Write Cycle Time	285		320		410		ns	
t _{RRW}	RMW Cycle $\overline{\text{RAS}}$ Pulse Width	165	10000	190	10000	265	10000	ns	
t _{CRW}	RMW Cycle $\overline{\text{CAS}}$ Pulse Width	105	10000	120	10000	185	10000	ns	
t _{RWD}	$\overline{\text{RAS}}$ to $\overline{\text{WE}}$ Delay	100		120		150		ns	9
t _{CWD}	$\overline{\text{CAS}}$ to $\overline{\text{WE}}$ Delay	55		65		80		ns	9

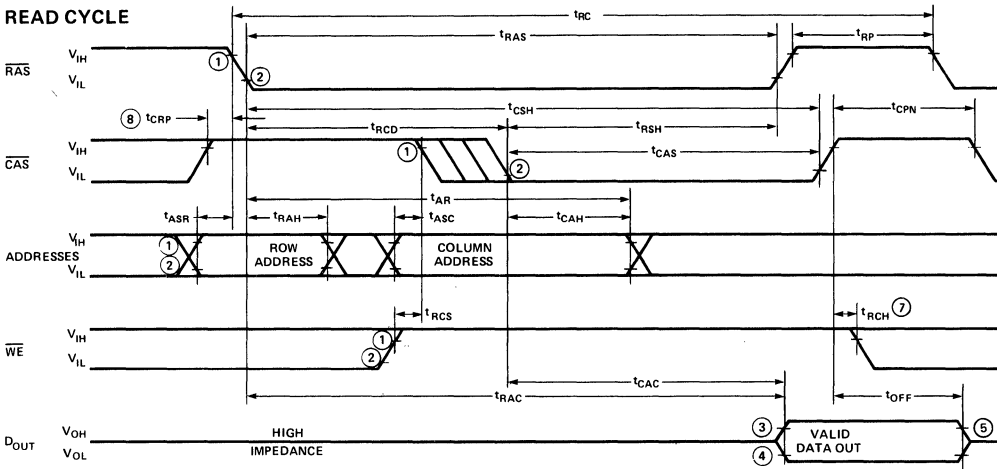
NOTES:

- All voltages referenced to V_{SS}.
- Eight cycles are required after power-up or prolonged periods (greater than 2ms) of $\overline{\text{RAS}}$ inactivity before proper device operation is achieved. Any 8 cycles which perform refresh are adequate for this purpose.
- A.C. Characteristics assume t_r = 5ns.
- Assume that t_{RCD} ≤ t_{RCD} (max.). If t_{RCD} is greater than t_{RCD} (max.) then t_{RAC} will increase by the amount that t_{RCD} exceeds t_{RCD} (max.).
- Load = 2 TTL loads and 100pF.
- Assumes t_{RCD} ≥ t_{RCD} (max.).

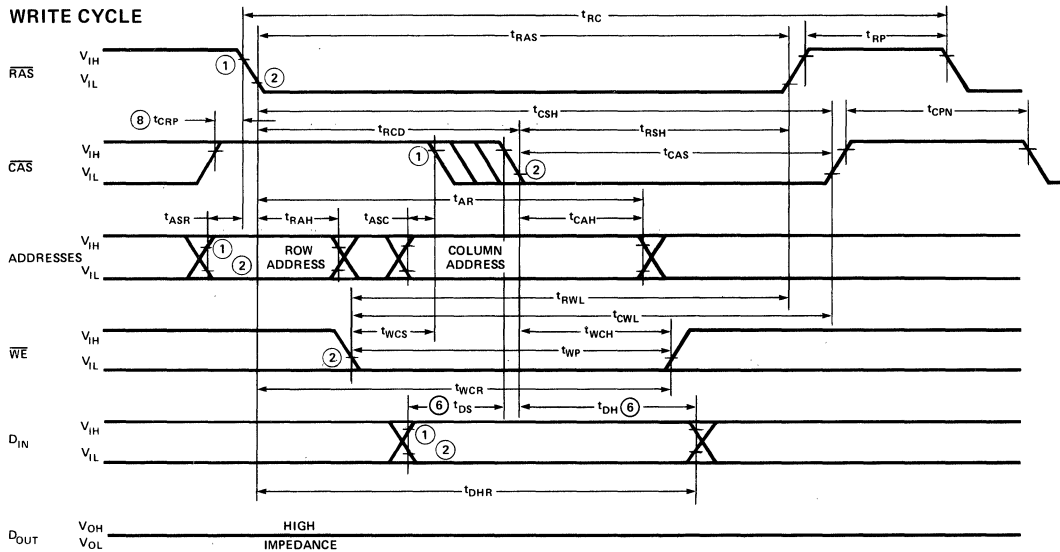
- t_{RCD} (max.) is specified as a reference point only; if t_{RCD} is less than t_{RCD} (max.) access time is t_{RAC}. If t_{RCD} is greater than t_{RCD} (max.) access time is t_{RCD} + t_{CAC}.
- t_T is measured between V_{IH} (min.) and V_{IL} (max.).
- t_{WCS}, t_{CWB} and t_{WPB} are specified as reference points only. If t_{WCS} ≥ t_{WCS} (min.) the cycle is an early write cycle and the data out pin will remain high impedance throughout the entire cycle. If t_{CWB} ≥ t_{CWB} (min.) and t_{WPB} ≥ t_{WPB} (min.), the cycle is a read-modify-write cycle and the data out will contain the data read from the selected address. If neither of the above conditions is satisfied, the condition of the data out is indeterminate.

WAVEFORMS

READ CYCLE



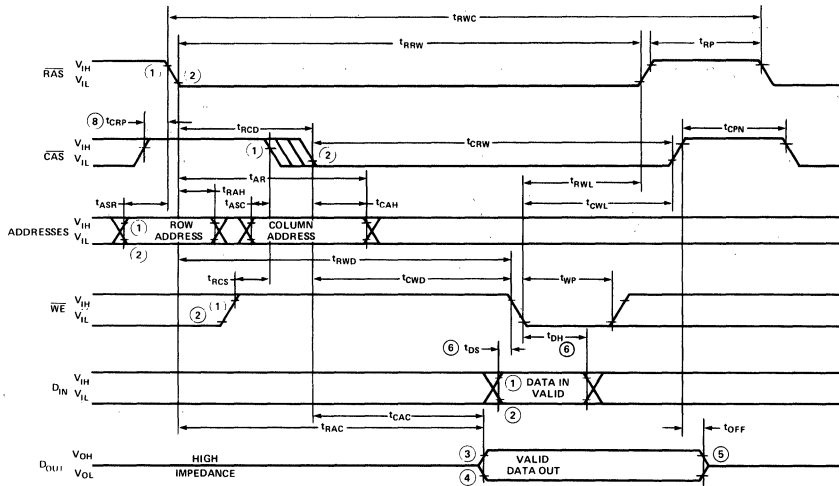
WRITE CYCLE



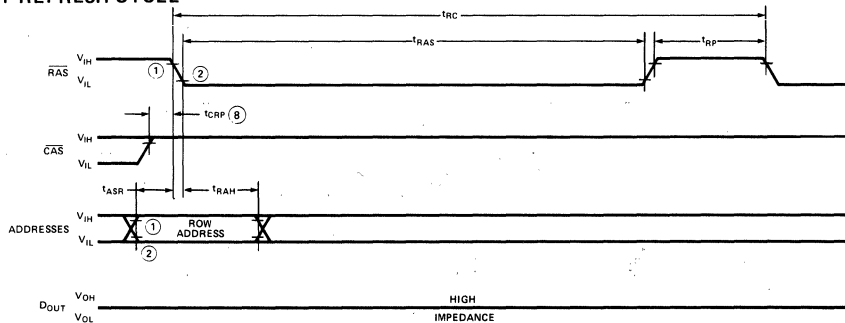
- NOTES:
- 1, 2. V_{IH} MIN AND V_{IL} MAX ARE REFERENCE LEVELS FOR MEASURING TIMING OF INPUT SIGNALS.
 - 3, 4. V_{OH} MIN AND V_{OL} MAX ARE REFERENCE LEVELS FOR MEASURING TIMING OF D_{OUT} .
 5. t_{OFF} IS MEASURED TO $I_{OUT} < |I_{OL}|$.
 6. t_{DS} AND t_{DH} ARE REFERENCED TO CAS OR WE , WHICHEVER OCCURS LAST.
 7. t_{RCH} IS REFERENCED TO THE TRAILING EDGE OF CAS OR RAS , WHICHEVER OCCURS FIRST.
 8. t_{CRP} REQUIREMENT IS ONLY APPLICABLE FOR RAS/CAS CYCLES PRECEDED BY A CAS -ONLY CYCLE (i.e., FOR SYSTEMS WHERE CAS HAS NOT BEEN DECODED WITH RAS).

WAVEFORMS

READ-MODIFY-WRITE CYCLE

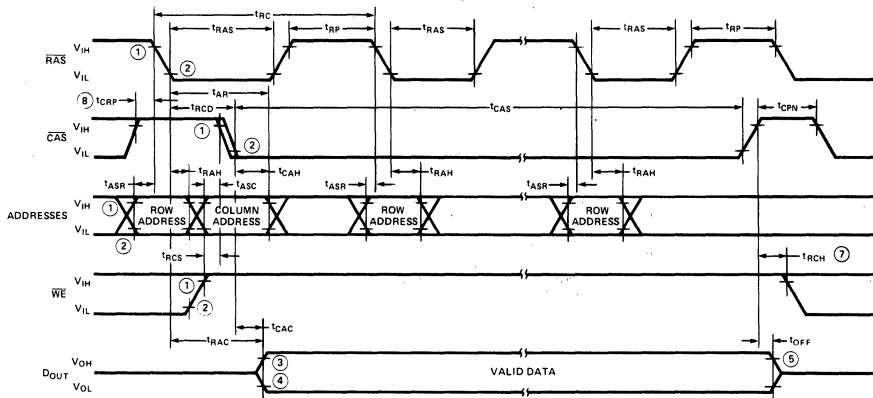


RAS-ONLY REFRESH CYCLE



HIDDEN REFRESH CYCLE

(For Hidden Refresh Operation order 2118-10 S6445, 2118-12 S6446 or 2118-15 S6447)



- NOTES: 1.2. V_{IH} MIN AND V_{IL} MAX ARE REFERENCE LEVELS FOR MEASURING TIMING OF INPUT SIGNALS.
 3.4. V_{OH} MIN AND V_{OL} MAX ARE REFERENCE LEVELS FOR MEASURING TIMING OF D_{OUT} .
 5. t_{OFF} IS MEASURED TO $I_{OUT} \leq |I_{OL}|$.
 6. t_{DS} AND t_{DH} ARE REFERENCED TO CAS OR WE , WHICHEVER OCCURS LAST.
 7. t_{RCH} IS REFERENCED TO THE TRAILING EDGE OF CAS OR RAS , WHICHEVER OCCURS FIRST.
 8. t_{CRP} REQUIREMENT IS ONLY APPLICABLE FOR RAS/CAS CYCLES PRECEDED BY A CAS -ONLY CYCLE (I.E., FOR SYSTEMS WHERE CAS HAS NOT BEEN DECODED WITH RAS).

2118 FAMILY

D.C. AND A.C. CHARACTERISTICS, PAGE MODE^[7,8,11]

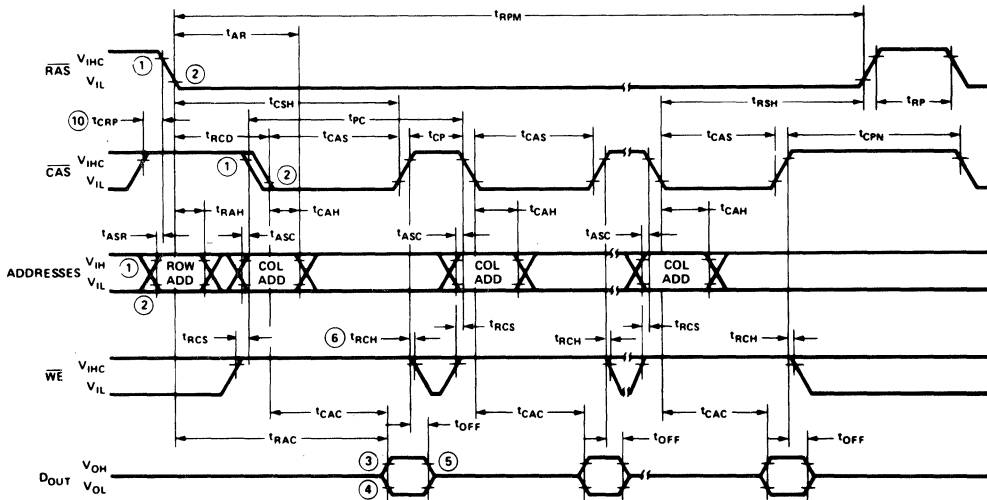
$T_A = 0^\circ\text{C}$ to 70°C , $V_{DD} = 5\text{V} \pm 10\%$, $V_{SS} = 0\text{V}$, unless otherwise noted.

(For Page Mode Operation order 2118-10 S6329, 2118-12 S6330 or 2118-15 S6331)

Symbol	Parameter	2118-10 S6329		2118-12 S6330		2118-15 S6331		Unit	Notes
		Min.	Max.	Min.	Max.	Min.	Max.		
t_{PC}	Page Mode Read or Write Cycle	125		145		190		ns	
t_{PCM}	Page Mode Read Modify Write Cycle	175		200		280		ns	
t_{CP}	$\overline{\text{CAS}}$ Precharge Time, Page Cycle	60		70		85		ns	
t_{RPM}	$\overline{\text{RAS}}$ Pulse Width, Page Mode	115	10000	140	10000	175	10000	ns	
t_{CAS}	$\overline{\text{CAS}}$ Pulse Width	55	10000	65	10000	95	10000	ns	
I_{DD4}	V_{DD} Supply Current Page Mode, Minimum t_{PC} , Minimum t_{CAS}		20		17		15	mA	

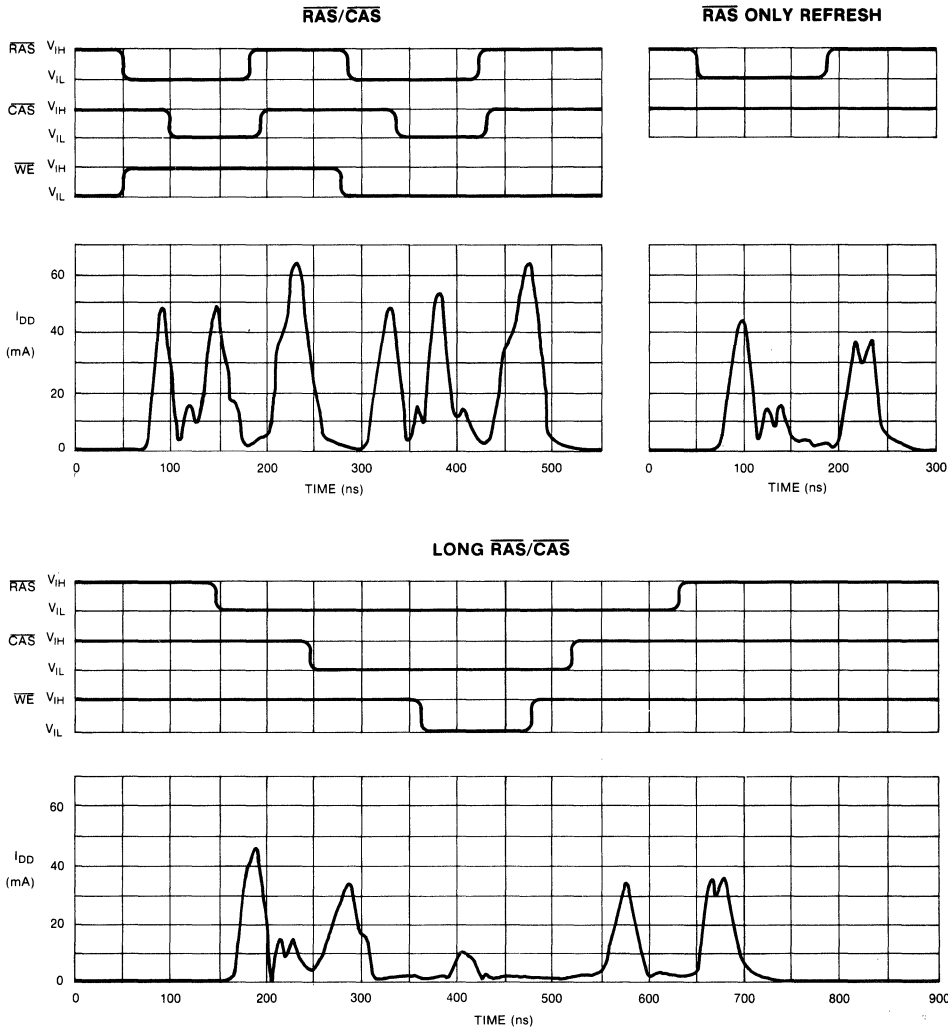
WAVEFORMS

PAGE MODE READ CYCLE



- NOTES:
1. $V_{IH\text{ MIN}}$ AND $V_{IL\text{ MAX}}$ ARE REFERENCE LEVELS FOR MEASURING TIMING OF INPUT SIGNALS.
 2. $V_{OH\text{ MIN}}$ AND $V_{OL\text{ MAX}}$ ARE REFERENCE LEVELS FOR MEASURING TIMING OF D_{OUT} .
 3. t_{OFF} IS MEASURED TO $I_{OUT} = I_{OL}$.
 4. t_{CRP} REQUIREMENT IS ONLY APPLICABLE FOR $\overline{\text{RAS}}$ / $\overline{\text{CAS}}$ CYCLES PRECEDED BY A $\overline{\text{CAS}}$ -ONLY CYCLE (I.E., FOR SYSTEMS WHERE $\overline{\text{CAS}}$ HAS NOT BEEN DECODED WITH $\overline{\text{RAS}}$).
 5. t_{RCH} IS REFERENCED TO THE TRAILING EDGE OF $\overline{\text{CAS}}$ OR $\overline{\text{RAS}}$, WHICHEVER OCCURS FIRST.
 6. ALL VOLTAGES REFERENCED TO V_{SS} .
 7. AC CHARACTERISTIC ASSUME $t_r = 5\text{ns}$.
 8. SEE THE TYPICAL CHARACTERISTICS SECTION FOR VALUES OF THIS PARAMETER UNDER ALTERNATE CONDITIONS.
 9. t_{CRP} REQUIREMENT IS ONLY APPLICABLE FOR $\overline{\text{RAS}}$ / $\overline{\text{CAS}}$ CYCLES PRECEDED BY A $\overline{\text{CAS}}$ -ONLY CYCLE (I.E., FOR SYSTEMS WHERE $\overline{\text{CAS}}$ HAS NOT BEEN DECODED WITH $\overline{\text{RAS}}$).
 10. ALL PREVIOUSLY SPECIFIED A.C. AND D.C. CHARACTERISTICS ARE APPLICABLE TO THEIR RESPECTIVE PAGE MODE DEVICE (I.E., 2118-10 S6329 WILL OPERATE AS A 2118-10).

TYPICAL SUPPLY CURRENT WAVEFORMS



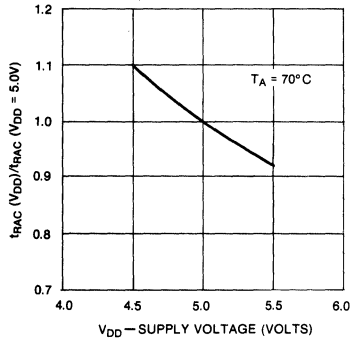
Typical power supply waveforms vs. time are shown for the $\overline{\text{RAS}}/\overline{\text{CAS}}$ timings of Read/Write, Read/Write (Long $\overline{\text{RAS}}/\overline{\text{CAS}}$), and $\overline{\text{RAS}}$ -only refresh cycles. I_{DD} current transients at the $\overline{\text{RAS}}$ and $\overline{\text{CAS}}$ edges require adequate decoupling of these supplies.

The effects of cycle time, V_{DD} supply voltage and ambient

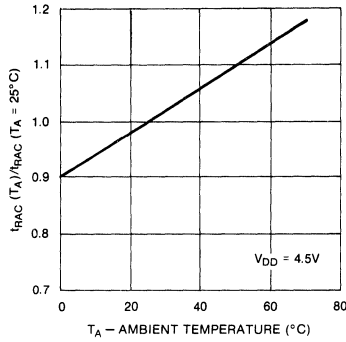
temperature on the I_{DD} current are shown in graphs included in the Typical Characteristics Section. Each family of curves for $I_{\text{DD}1}$, $I_{\text{DD}2}$, and $I_{\text{DD}3}$ is related by a common point at $V_{\text{DD}} = 5.0\text{V}$ and $T_A = 25^\circ\text{C}$ for two given t_{RAS} pulse widths. The typical I_{DD} current for a given condition of cycle time, V_{DD} and T_A can be determined by combining the effects of the appropriate family of curves.

TYPICAL CHARACTERISTICS

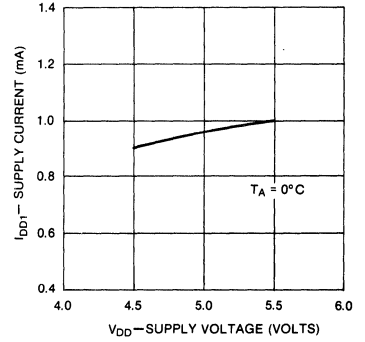
GRAPH 1
TYPICAL ACCESS TIME
 t_{RAC} (NORMALIZED) VS. V_{DD}



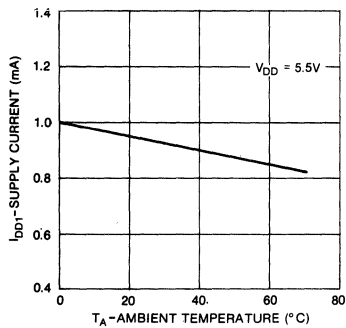
GRAPH 2
TYPICAL ACCESS TIME
 t_{RAC} (NORMALIZED) VS.
AMBIENT TEMPERATURE



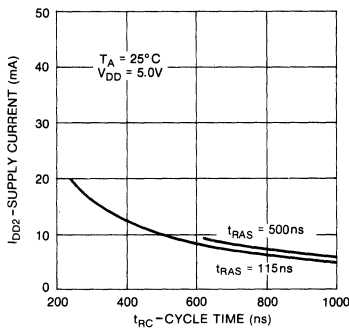
GRAPH 3
TYPICAL STANDBY CURRENT
 I_{DD1} VS. V_{DD}



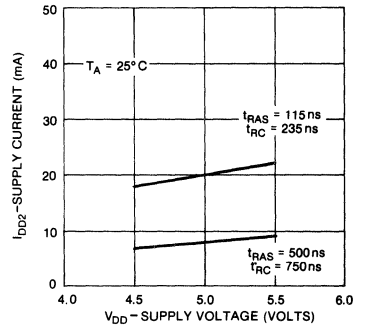
GRAPH 4
TYPICAL STANDBY CURRENT
 I_{DD1} VS. AMBIENT TEMPERATURE



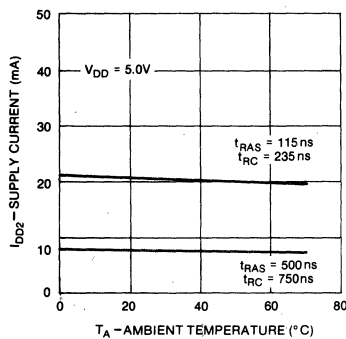
GRAPH 5
TYPICAL OPERATING CURRENT
 I_{DD2} VS. t_{RC}



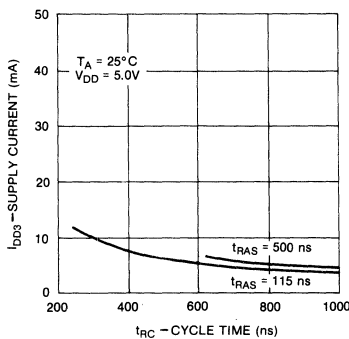
GRAPH 6
TYPICAL OPERATING CURRENT
 I_{DD2} VS. V_{DD}



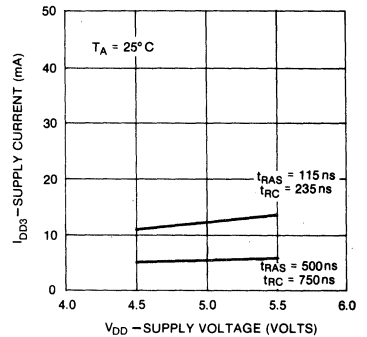
GRAPH 7
TYPICAL OPERATING CURRENT
 I_{DD2} VS. AMBIENT TEMPERATURE



GRAPH 8
TYPICAL RAS ONLY
REFRESH CURRENT
 I_{DD3} VS. t_{RC}

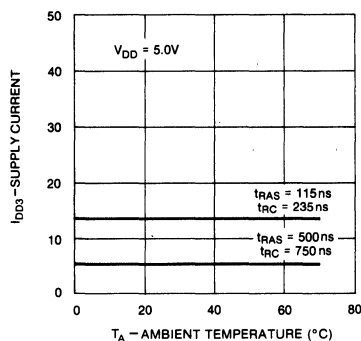


GRAPH 9
TYPICAL RAS ONLY
REFRESH CYCLE
 I_{DD3} VS. V_{DD}

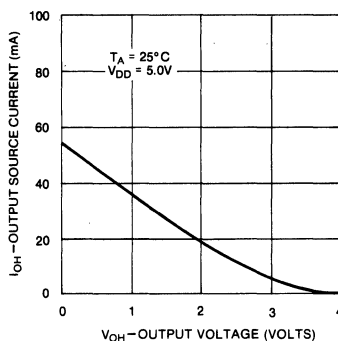


TYPICAL CHARACTERISTICS

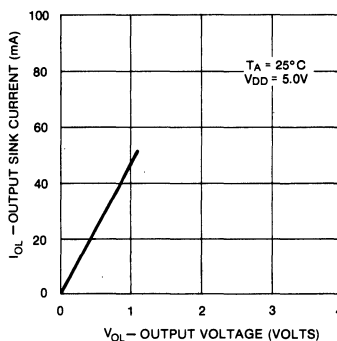
GRAPH 10
TYPICAL $\overline{\text{RAS}}$ ONLY
REFRESH CURRENT
 I_{DD3} VS. AMBIENT TEMPERATURE



GRAPH 11
TYPICAL OUTPUT SOURCE CURRENT
 I_{OH} VS. OUTPUT VOLTAGE V_{OH}



GRAPH 12
TYPICAL OUTPUT SINK CURRENT
 I_{OL} VS. OUTPUT VOLTAGE V_{OL}



DEVICE DESCRIPTION

The Intel® 2118 is produced with HMOS, a high performance MOS technology which incorporates on chip substrate bias generation. This process, combined with new circuit design concepts, allows the 2118 to operate from a single +5V power supply, eliminating the +12V and -5V requirements. Pins 1 and 9 are not connected, which allows P.C.B. layout for future higher density memory generations.

The 2118 is functionally compatible with the industry standard 16-pin 16K dynamic RAMs, except for the power supply requirements. Replacing the +12V supply with a +5V supply and eliminating the -5V bias altogether, allows simple upgrade both in power and performance. To achieve total speed performance upgrade, however, the timing circuitry must be modified to accommodate the higher performance.

READ CYCLE

A Read cycle is performed by maintaining Write Enable ($\overline{\text{WE}}$) high during a $\overline{\text{RAS}}$ /CAS operation. The output pin of a selected device will remain in a high impedance state until valid data appears at the output at access time.

Device access time, t_{ACC} , is the longer of the two calculated intervals:

$$1. t_{ACC} = t_{RAC} \quad \text{OR} \quad 2. t_{ACC} = t_{RCD} + t_{CAC}$$

Access time from $\overline{\text{RAS}}$, t_{RAC} , and access time from $\overline{\text{CAS}}$, t_{CAC} , are device parameters. Row to column address strobe delay time, t_{RCD} , are system dependent timing

parameters. For example, substituting the device parameters of the 2118-3 yields:

$$3. t_{ACC} = t_{RAC} = 100\text{nsec for } 25\text{nsec} \leq t_{RCD} \leq 45\text{nsec}$$

OR

$$4. t_{ACC} = t_{RCD} + t_{CAC} = t_{RCD} + 55\text{nsec for } t_{RCD} > 45\text{nsec}$$

Note that if $25\text{nsec} \leq t_{RCD} \leq 45\text{nsec}$ device access time is determined by equation 3 and is equal to t_{RAC} . If $t_{RCD} > 45\text{nsec}$ access time is determined by equation 4. This 20nsec interval (shown in the t_{RCD} inequality in equation 3) in which the falling edge of $\overline{\text{CAS}}$ can occur without affecting access time is provided to allow for system timing skew in the generation of $\overline{\text{CAS}}$.

REFRESH CYCLES

Each of the 128 rows of the 2118 must be refreshed every 2 milliseconds to maintain data. Any memory cycle:

1. Read Cycle
2. Write Cycle (Early Write, Delayed Write or Read-Modify-Write)
3. $\overline{\text{RAS}}$ -only Cycle

refreshes the selected row as defined by the low order ($\overline{\text{RAS}}$) addresses. Any Write cycle, of course, may change the state of the selected cell. Using a Read, Write, or Read-Modify-Write cycle for refresh is not recommended for systems which utilize "wire-OR" outputs since output bus contention will occur.

A $\overline{\text{RAS}}$ -only refresh cycle is the recommended technique for most applications to provide for data retention. A $\overline{\text{RAS}}$ -only refresh cycle maintains the $\overline{\text{DOUT}}$ in the high

impedance state with a typical power reduction of 30% over a Read or Write cycle.

RAS/CAS TIMING

$\overline{\text{RAS}}$ and $\overline{\text{CAS}}$ have minimum pulse widths as defined by t_{RAS} and t_{CAS} respectively. These minimum pulse widths must be maintained for proper device operation and data integrity. A cycle, once begun by bringing $\overline{\text{RAS}}$ and/or $\overline{\text{CAS}}$ low must not be ended or aborted prior to fulfilling the minimum clock signal pulse width(s). A new cycle can not begin until the minimum precharge time, t_{RP} , has been met.

DATA OUTPUT OPERATION

The 2118 Data Output (D_{OUT}), which has three-state capability, is controlled by $\overline{\text{CAS}}$. During $\overline{\text{CAS}}$ high state ($\overline{\text{CAS}}$ at V_{IH}) the output is in the high impedance state. The following table summarizes the D_{OUT} state for various types of cycles.

Intel 2118 Data Output Operation for Various Types of Cycles

Type of Cycle	D_{OUT} State
Read Cycle	Data From Addressed Memory Cell
Early Write Cycle	HI-Z
RAS-Only Refresh Cycle	HI-Z
CAS-Only Cycle	HI-Z
Read/Modify/Write Cycle	Data From Addressed Memory Cell
Delayed Write Cycle	Indeterminate

HIDDEN REFRESH

An optional feature of the 2118 is that refresh cycles may be performed while maintaining valid data at the output pin. This feature is referred to as Hidden Refresh. Hidden Refresh is performed by holding $\overline{\text{CAS}}$ at V_{IL} and taking $\overline{\text{RAS}}$ high and after a specified precharge period (t_{RP}), executing a "RAS-Only" refresh cycle, but with $\overline{\text{CAS}}$ held low (see Figure 1).

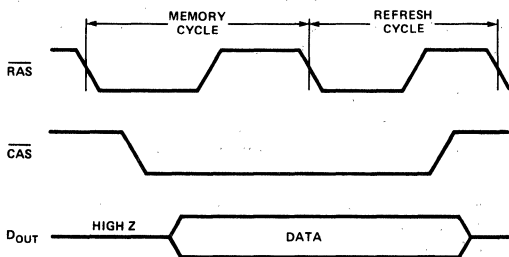


Figure 1. Hidden Refresh Cycle.

This feature allows a refresh cycle to be "hidden" among data cycles without affecting the data availability.

POWER ON

After the application of the V_{DD} supply, or after extended periods of bias (greater than 2ms) without clocks, the device must perform a minimum of eight (8) initialization cycles (any combination of cycles containing a $\overline{\text{RAS}}$ clock such as $\overline{\text{RAS}}$ -only refresh) prior to normal operation.

The V_{DD} current (I_{DD}) requirement of the 2118 during power on is, however, dependent upon the input levels of $\overline{\text{RAS}}$ and $\overline{\text{CAS}}$. If the input levels of these clocks are at V_{IH} or V_{DD} , whichever is lower, the I_{DD} requirement per device is I_{DD1} (I_{DD} standby). If the input levels for these clocks are lower than V_{IH} or V_{DD} the I_{DD} requirement will be greater than I_{DD1} , as shown in Figure 2.

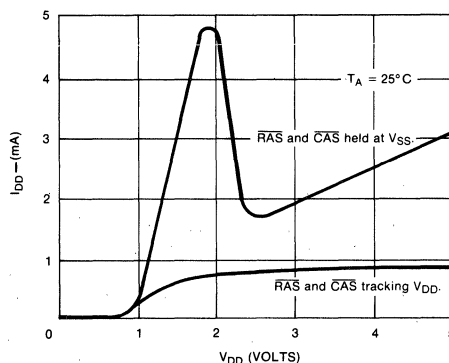


Figure 2. Typical I_{DD} VS V_{DD} during power up.

For large systems, this current requirement for I_{DD} could be substantially more than that for which the system has been designed. A system which has been designed, assuming the majority of devices to be operating in the refresh/standby mode, may produce sufficient I_{DD} loading such that the power supply may current limit. To assure that the system will not experience such loading during power on, a pullup resistor for each clock input to V_{DD} to maintain the non-selected current level (I_{DD1}) for the power supply is recommended.